

Dialect Dive: The Intersection Between Linguistics and Technology

CSC-475: Seminar in Computer Science
Spring 2024

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Abstract

Following Mandarin, Spanish is one of the most spoken languages in the world. Over time Spanish has slowly diverged into many different regional variations. There are Catalan, Andalusian, Caribbean, and Andean, to name a few. There is a plethora of information and research surrounding the language's diversity. However, studying Spanish and its variations can be difficult because this information is stratified across the internet. This is where Dialect Dive came to fruition. This website is aimed towards the second language learner, also commonly referred to as L2 learner. This website is an online resource that provides information on these regional variations within Spanish and the distinct phonemes of each variety. Honing in on these phonemes helps L2 learners understand the key distinguishing factors of these variations, ultimately supporting the growth of their listening and speaking skills. The website is accessible at <https://ainsleyyoshizumi.github.io/CSC475Dialects/website/> and features a map and a chart of the International phonetic alphabet at the bottom of the page. The map shows the languages that have a given phoneme and their location in the world.

1 Introduction

Today, Spanish is spoken on four continents including the Americas, Europe, Africa, and Asia. Over time Spanish has diverged into many different dialects, indigenous and non-indigenous. Some popular dialects in Spain include Catalan and Castilian. In Latin America, there is Mexican Spanish, Argentinian Spanish, Chilean Spanish, the list goes on and on. A common misconception when it comes to Spanish speakers is one considering Spanish as the sole language or dialect being spoken. However, Spanish is incredibly diverse. Studying Spanish and its morphology can unlock many new insights and perspectives on the culture and sounds of the language. [1] [2] [3]

Spanish accents can be identified by two phonetic traits, suprasegmental and segmental. Segmental refers to vowels and consonants and suprasegmental refers to other speech attributes that accompany the segmental traits. These variations are caused by a variety of different variables, including political boundaries, indigenous populations, isolation, geographic location, and European settlement. Studying these variations of Spanish reveals important information regarding the diversity within a language and subtle nuances within a culture, that can vary region by region. [2] [3]

Studying these dialectical variations can further uncover fascinating insights into the social, historical, and cultural dynamics of Spanish-speaking communities. These dialects not only reflect linguistic diversity but also serve as markers of identity and heritage for those who speak them. Investigating the nuances of Spanish dialects, researchers gain a deeper understanding of how language evolves and adapts within different contexts, shedding light on the intricacies between language, society, and human interaction.

The database Dialect Dive was created with comes from PHOIBLE. PHOIBLE serves as a collection of phonological inventory data across languages, gathered from source documents and tertiary databases and consolidated into a unified, easily searchable dataset. The 2019 Release 2.0 comprises 3020 inventories, encompassing 3183 segment types identified in 2186 different languages.

Additionally, part of this project encompasses creating an audio classifier that would be able to identify different Spanish dialects. The classifier was trained on a crowd-sourced dataset created by Google language resources found on <https://www.openslr.org/71> and <https://www.openslr.org/72>. At the current moment, the classifier is only trained on Chilean and Colombian female audios. Using Python, we took the audio files from this dataset created another dataset containing y-vector values, and trained the audio classifier on this. [4]

2 Related Work

Spanish phonology is an area of study with a plethora of information. There are documented 19 consonant sounds amongst the 8 listed countries of Cuba, Dominican Republic, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, and Puerto Rico. This is shown in figure 1, which is taken from article [3]. The screenshot only shows a portion of consonant sounds within the language, it is also important to note that there very well may exist other undocumented variations outside of Spanish phonological literature. Since these phonological variations between Spanish dialects have been identified and explained, it is important to also mention that additional factors like contextual, situational, sociolinguistic, and word positioning also govern allophonic realizations. Furthermore, while there is discourse surrounding Spanish phonology, there is still more research that must be undertaken to learn more about how dialects can affect language acquisition.

Listening comprehension is a particular area of interest in this project as this program aims to serve as a tool to support students in this area of language acquisition. By definition, listening is simply viewed as the act of paying attention, particularly to sound. However, the actual meaning of listening should be regarded as a process. Teachers operate under the fallacy that their students know how to listen. A combination of actions must transpire for listening to formally occur, listening is like all other skills, and the assumption that listening will develop as students increase their proficiency limits the impact teachers can have on their students [5].

A study performed by Christine Goh found some difficulties that occur when students attempt to listen to native speakers are failure to recognize vocabulary, missing the input because they were too busy trying to understand, and inability to chunk the stream of language properly to interpret it [6]. Comprehension plays a pivotal role in the communication process [7], a study on phonological perception between L1 and L2 learners found confirmation that for pedagogical purposes, a bottom-up approach to second language acquisition proved to be the most influential.

The study revealed that bottom-up is stimulus-driven, meaning audibly hearing language streams provoke bottom-up learning in students. Moreover, learners require precise bottom-up skills to effectively contrast incoming information with the hypotheses and anticipations formed during top-down processing [8]. Honing in on bottom-up skills becomes indispensable for learners, enabling them to take in audio and enmesh it with sensory preexisting knowledge frameworks, thereby fostering deeper comprehension and retention of the language.

In a meta-analysis of the effect of mobile-assisted learning apps, researchers found through the analysis of 17 total qualitative and quantitative studies that mobile-assisted learning apps are in general, beneficial for those learning a second language. However, there were limitations involving the motivations of the users of these apps. This means the success of the user in learning another language is dependent on many individual factors regarding desire, goals, and personal motivations for learning a second language. With this information in mind, constructing language acquisition-supporting websites and apps like Dialect Dive can only aid in the support of L2 learners and their processes. Having information organized and readily available for users draws the attention of students and educators because of the convenience associated with this organization. Dialect Dive conveniently aggregating this information in a comprehensible format closes many gaps that can help make second language acquisition an easier process for learners. [9]

The research and discourse surrounding mobile-assisted language learning is new and something that still requires extensive research. The addition of this Spanish dialect recognition program will add to

Phoneme	Allophonic realization	Cuba	Dominican Republic	El Salvador	Guatemala	Honduras	Mexico	Nicaragua	Puerto Rico
Plosives									
/p/	[b]								
	[k]								
/b/	[β]								
	[v]								
	[g]*								
/t/	[t]								
/d/	[ð]								
	[ø]								
	[r]								
/k/	[g]								
/g/	[ø]								
	[χ]								
	[ŋ]								

Figure 1: Martinez 2011

the discourse surrounding the topic of MALL and can be used in conjunction with other research, creating more concrete support for the topic. This program will be an effective tool for L2 learners as well, supporting their education and listening comprehension skills.

Students using this tool will be exposed to different perspectives of the Spanish language, including phonetic variance, lexicon, morphology, and grammatical structures. By training and testing the L2 student's ears, their skills to naturally parse and comprehend native speech will increase, benefiting not only their level of Spanish but their overall confidence in themselves as an L2 student as well.

3 Data

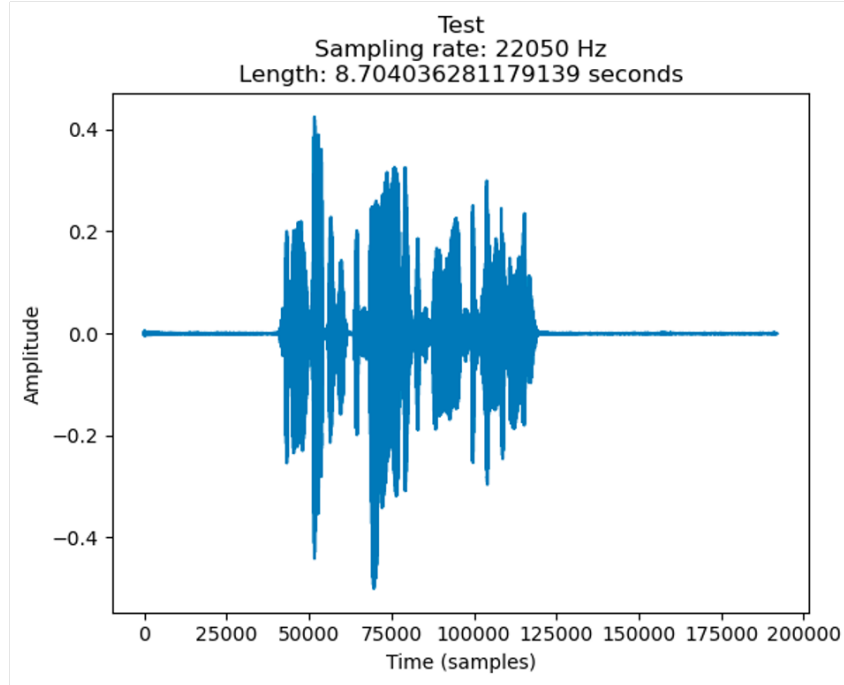


Figure 2: Amplitude graph of one audio file

Audio Filename	V_0	V_1	V_2	V_3	V_4
File 1	-0.000106	-0.000219	-0.000209	-0.000246	-0.000185
File 2	0.00097	0.001404	0.001253	0.001405	0.001204
File 3	-0.000174	-0.000079	0.000005	-0.000186	-0.000086
File 4	0.000069	0.000033	0.000045	0.000058	0.000038
File 5	0.000002	0.000018	-0.000014	0.000015	0.000021

Figure 3: Aggregated y-vector values

The dataset used to create Dialect Dive comes from PHOIBLE, abbreviated from "Phonetics Information Base and Lexicon," which acts as a thorough database containing phonological inventory data spanning various languages worldwide. Its main objective is to compile information on the sounds, known as phonemes, employed across diverse linguistic contexts. This data is sourced from a range of materials, such as linguistic documents and other databases, and then consolidated into a cohesive, searchable dataset. PHOIBLE serves as a vital tool for linguists, researchers, and language enthusiasts interested in delving into the intricate web of phonological diversity exhibited across different languages globally. Because of its brevity, PHOIBLE served to be the perfect database to base Dialect Dive off of.

PHOIBLE is created in collaboration with other contributors such as UCLA Phonological Segment Inventory Database and the Stanford Phonology Archive. Additionally, there is information in the database from phonological inventories in other countries like South America. This contributes to a more diverse and inclusive dataset as it also includes information about indigenous languages.

The other dataset used in this project to create the classifier from was found in Google language resources. This repository contains audio datasets from 31 different languages, separating them by male and female datasets. Furthermore, there are datasets for multiple dialects of Spanish including, Chilean, Colombian, Venezuelan, and Puerto Rican, Peruvian, and Catalan. The Chilean and Colombian female datasets were used for the classifier, each containing well over a thousand different audio .WAV files. These files were then uploaded into virtual studio code where we used python to construct graphs for each of the audio files using time and amplitude for the X and Y-axis' respectively. This can be seen in figure 2 below.

These graphs produced y-vector values which could then be used to train the data, the graph in figure 3 below shows a small portion of the aggregated data produced from these graphs.

4 Methods

Pictured below in figure 4 is a chart depicting the process of this project. Dialect Dive can be broken down into two separate parts, the first being the speech classifier and the second being the phonetic map of world languages. Starting with the speech classifier, as mentioned earlier, the classifier was built and trained on Chilean and Colombian female .WAV audios. In order to format this .WAV file so the computer could process it, we used Librosa, a Python package for music and audio analysis. We also used the very popular data analysis library Pandas for data manipulations. We used code that went through each line in the .CSV file created for each aggregated data file (Chilean and Colombian) and classified the y-vector values as a 1 if the .WAV file was Chilean or 0 if it was Colombian. As mentioned in figure 2, we took this aggregated data file and ran a K-Nearest Neighbor test on it.

We additionally played around with the idea of taking user input and feeding it into the classifier to see how our model would perform. We ran sample tests with audio from the dataset and audio that contained no Spanish spoken in it. Through these tests, we ran into the issue of our inability to format the data properly to match the audio on which the classifier was trained. This forced us to pivot to the bottom half of the process pipeline, which was creating a visualization of the phonemes across world languages.

In the bottom half of the chart is the interface portion of Dialect Dive's map. The map is based on phonemes found in the international phonetic alphabet and dialectical distinctions across Spanish varieties. The IPA chart and dialectical distinctions are reliant on each other as the phonemes shown on the website's IPA chart are relative only to Spanish linguistics. These phonemes are important for making distinctions between varieties of Spanish and also relate to a speaker's ability to identify the specific type of dialect being spoken. We chose these specific phonemes based on the scholarly research mentioned earlier in the related work section [10] [3] and PHOIBLE's dataset as well [11]

We chose to break down the map through IPA charts and distinctions because it made more sense to organize the data in that way. PHOIBLE was a pre-established dataset so we did not apply any pre-processing to it and kept all attributes within the dataset the same. The IPA chart is slightly edited to account for the focus of this project being on the Spanish language and the IPA chart is not a direct reflection of the full IPA.

The repository to which one can access the code for Dialect Dive and the classifier can be found at <https://github.com/ainsleyyoshizumi/CSC475dialects/>.

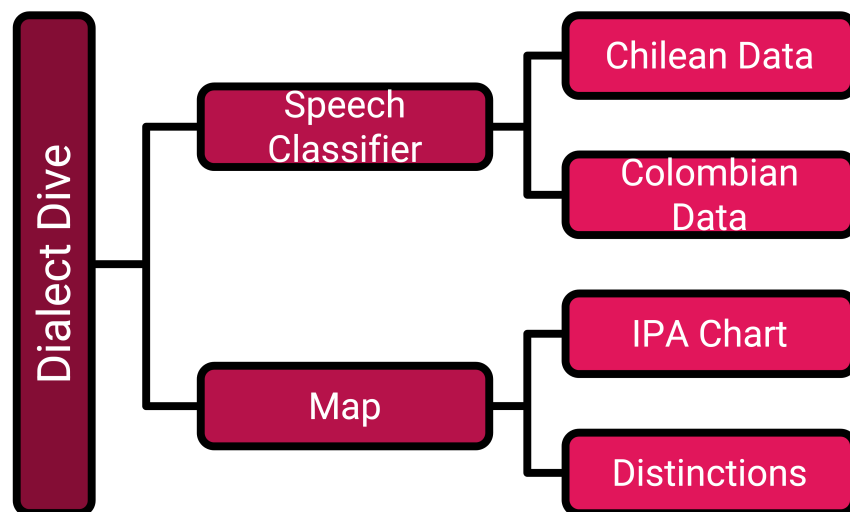


Figure 4: Pipeline of the project

5 Results

5.1 Dialect Dive

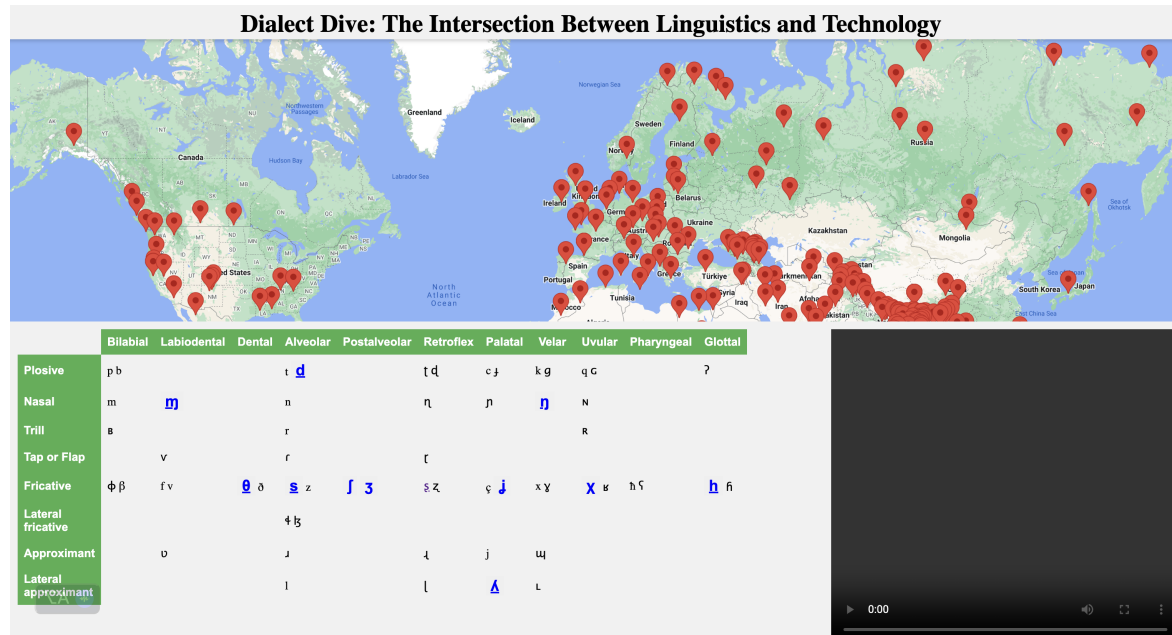


Figure 5: Dialect Dive

Figure 5 is a picture of what Dialect Dive currently looks like. The website opens to a map of the world with various markers showing the different locations of world languages. In the bottom half of the screen is the International Phonetic Alphabet chart. The chart shows the many different symbols for each consonant in the International Phonetic Alphabet. The consonants that are highlighted contain links to videos illustrating two things, what the consonant sounds like and how to produce the sound.

Understanding how to accurately pronounce consonants in the International Phonetic Alphabet (IPA) holds significant importance for linguists, language learners, and speech professionals alike. Consonants play a crucial role in shaping the sounds of speech. Mastery of consonant pronunciation in the IPA allows for precise phonetic transcription, enabling researchers and students to accurately document and analyze the sounds of different languages. For language learners, grasping consonant articulation in the IPA facilitates the acquisition of new languages by providing a standardized reference point for pronunciation. Overall, a thorough understanding of consonant pronunciation in the IPA serves as a foundational skill in effective communication and language learning processes.

[1] [3]

5.2 Classifier

Figure 6 is a results table from the K-NN test run on the Chilean and Colombian data. The highest accuracy score we were able to get the model to perform at was only sixty-nine percent. This is a great limitation of the classifier built, it is important to mention the amount of data input to the K-NN test. In total, there were at least three gigabytes of y-vector values that the model processed. This likely impacted the accuracy of the classifier because of the plethora of redundant values in the dataset.

Regional Variety	Precision	Recall	f1-score	Support
Colombian	0.66	0.96	0.78	472
Chilean	0.86	0.33	0.47	350
Accuracy			0.69	822
Macro Avg.	0.76	0.64	0.63	822
Weighted Avg.	0.74	0.69	0.65	822

Figure 6: Results table

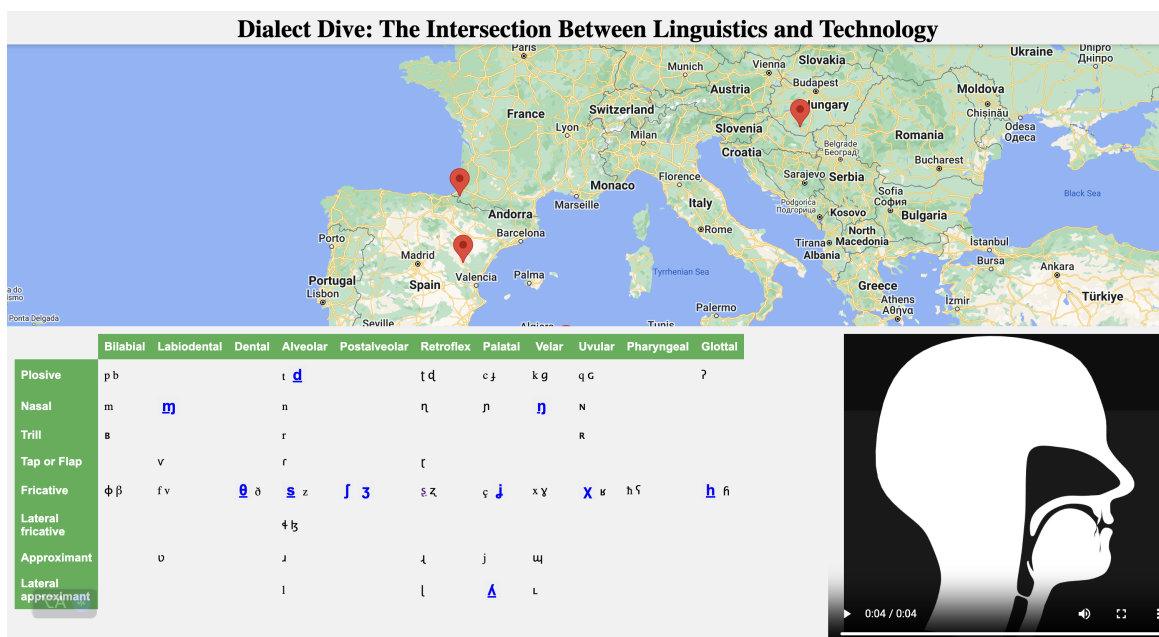


Figure 7: Dialect Dive

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

(a) Figure8 : DialectDive'sIPAChart

CONSONANTS (PULMONIC)											
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

(b) Figure9 : IPAChart

5.3 IPA Chart

Figure 7 depicts what happens when a user interacts with Dialect Dive by clicking on a linked phoneme. The phonemes link to videos that depict how to produce the consonant sound. The phonemes highlighted are ones of particular interest to the Spanish language. These phonemes are important for the distinction of various Spanish dialects because they represent a discrete unit of sound that can differentiate meaning within language.

Figure 8 and 9 show a comparison between the IPA chart and Dialect Dive's IPA chart which we created. It contains many of the same features, the only difference is that Dialect Dive only contains links for some of the constants in the chart as opposed to the chart found on Seeing Sounds, which contains a link for every pulmonic consonant. Pulmonic consonants, essential in speech production, entail the flow of air through the respiratory system and vocal tract, commonly governed by the movements of the lips, tongue, and palate. Their characteristics are discerned by how they are produced, where they are produced, and whether they involve vocal cord vibration. This is how the IPA chart is organized, as one can see, the x-axis of the graph lists the location in the mouth in which the consonant is produced and the y-axis is the mode or nature by which the sound is made in the mouth. As mentioned before, this is because of the focus this project has on the Spanish language and its consonants. [12]

6 Future Work

Dialect Dive is not by any means a completed project. There are limitations regarding the depth to which the dataset from PHOIBLE goes within different Spanish dialects. As of right now, the database only has roughly three Spanish dialects and they are all dialects from the Iberian Peninsula. There needs to be more work done regarding Spanish dialects in Latin America. The necessity for an extensive dataset focusing on Spanish dialects in Latin America stems from the region's linguistic diversity.

Latin America boasts a wide array of Spanish dialects, each influenced by unique historical, cultural, and geographical factors specific to its area. Despite Spanish's global presence and popularity, there's a scarcity of resources tailored to cataloging and analyzing these dialectical variations. Establishing a dedicated dataset for Spanish dialects in Latin America would not only boost linguistic research but also offer valuable insights into the sociolinguistic dynamics of the region. This dataset could help catalog existing patterns of linguistic variation, offer insight into languages in contact, and monitor linguistic heritage. Furthermore, it could serve as a vital tool for language learners and educators aiming to foster linguistic diversity and cultural appreciation within Spanish-speaking communities across Latin America.

Other features that would add to Dialect Dive would be adding map features based on region. Giving users the ability to explore a region and the phonemes found within a language in that specific region would shed light on other linguistic phenomena like morphology and lexicology. A big inspiration for this project is owed to the University of North Carolina State's Library of Spanish Voices [?]. This online website is a collection of student-sourced audio data organized via a map. This educational resource illustrates variational differences between Spanish across Latin America, Spain, and North Carolina. It also allows users to compare and contrast different native Spanish speakers' audio and includes transcripts for the audio in English and Spanish. The convenience and straightforwardness of this website provided the groundwork on which to base Dialect Dive.

Additionally, there is still work that must be done on the classifier constructed from the Chilean and Colombian datasets. There are hopes to have the classifier one day extend to multiple dialects. A program that recognizes Spanish dialects is possible, but there are still limitations regarding formatting the audio files and the computer's ability to extract key information like acoustic properties, phonetic patterns, and prosodic elements from the audio data.

In the future, this work could be used as a key tool for educators and L2 learners alike. Although there are limitations to this research, Dialect Dive still holds as an educational tool that can be used to explore phonemes and languages across the world.

References

- [1] Jose Ignacio Hualde. *Introduccion a la linguistica hispanica*. Cambridge Univ. Press, 2011.
- [2] Joseph A Wiczorek. Spanish dialects and the foreign language textbook: A sound perspective. *Hispania*, 74(1):175–181, 1991.
- [3] Silvia Martinez. Dialectal variations in spanish phonology: a literature review. *Echo*, 6(2):1–8, 2011.
- [4] Multiple Contributors. Language-Resource Repository. [://github.com/google/language-resources?tab=readme-ov-file](https://github.com/google/language-resources?tab=readme-ov-file). [Online; accessed 24-January-2023].
- [5] Steven Brown. *Listening myths: Applying second language research to classroom teaching*. University of Michigan Press, 2011.
- [6] Christine C.M Goh. A cognitive perspective on language learners' listening comprehension problems. *System*, 28(1):55–75, March 2000.
- [7] Joseph Siegel and Aki Siegel. Getting to the bottom of l2 listening instruction: Making a case for bottom-up activities. *Studies in Second Language Learning and Teaching*, 5:637–662, 2015.
- [8] Khaled Moradi. The impact of listening strategy instruction on academic lecture comprehension: A case of iranian efl learners. *Procedia - Social and Behavioral Sciences*, 70:406–416, 2013. Akdeniz Language Studies Conference, May, 2012, Turkey.

- [9] Sandra Figueiredo. The effect of mobile-assisted learning in real language attainment: A systematic review. *Journal of Computer Assisted Learning*, 39(4):1083–1102, 2023.
- [10] CHRISTINE SHEA and JEFFREY RENAUD. L2 perception of spanish palatal variants across different tasks. *Bilingualism: Language and Cognition*, 17(1):203–221, 2014.
- [11]
- [12] Seeing Speech: IPA Charts. <https://seeingspeech.ac.uk/ipa-charts/?chart=1>. Seeing Speech: IPA Charts.